R V College of Engineering
Department of Computer Science and Engineering
CIE - II: Question Paper

Subject : (Code)	**Database Management Systems (CD252IA)**	**Semester :**	**5TH BE**
Date :07/01/2025	**Duration :** 120 minutes	**Staff :HR/CNS/PD/SB/SNM/PH/MNV/PT**	
Name :	**USN :**	**Section :**	**CS-A/B/C/D/CD/CY/IS/AIML**

S.N	PART-A	M	BT	Co
1.1	Consider the following tables T1 and T2 , show the result of the operation T1 ⋈ (T1.P = T2.A AND T1.R = T2.C) T2 T1 P Q R 10 A 5 15 B 8 25 A 6 T2 A B C 10 b 6 25 c 3 10 b 5 P R 5 5	2	L3	4
1.2.	Write the 2 rules which must be satisfied , if a set of attributes FK in relation schema R1 is a foreign key of R1 that references relation R2	2	L1	2
1.3	Give an example for EXISTS operator of SQL(Considering the example of Insurance Database Schema of question no. 4 of PART-B)	2	L2	3
1.4	Write reflexive, decomposition/projective inference rules for functional dependencies $x \to y$ $x \to yz$	2	L1	1
1.5	What is a correlated query? Give example.	2	L2	4
	PART-B			
2 a.	Explain the following relational model constraints with example. i Domain Constraint ii Key Constraint iii Entity Integrity Constraint	6	L1	2
b.	Explain how the relational model constraints that may be violated by delete operation. and the types of actions that may be taken if delete operation causes a violation	4	L2	3
3 a.	Explain the relational algebra operation for set theory with examples.	6	L1	1
b.	Explain DIVISION operation of relational algebra with an example.	4	L2	4
4	Consider the Insurance Database given below. PERSON (driver-id#:, name , address) CAR (Regno, model, year) ACCIDENT (Report-Number, date, location) OWNS (driver-id#, Regno) PARTICIPATED (driver-id#, Regno,Report_number, damageamount) Write the queries in relational algebra to: i Find driver-id# of every person, who owns a 'Toyota Fortuner' or a 'Hyundai Creta' car model ii Find the driver-id#, name of every person who has ever been involved in some car accident iii Find the number of accidents in which cars belonging to each model were	10	L3	4

	involved iv Find the driver-id# and name of all persons who have had all of their cars involved in some accident			
5.	Consider the following Schema : Sailors(sid: integer, sname: string, rating: integer, age: real); Boats(bid: integer, bname: string, color: string); Reserves(sid: integer, bid: integer, day: date) Write the queries in SQL to: i Find the sailors information whose name begins and ends with 'A' and has at least 3 characters. ii Find the ids of sailors who have reserved a red boat or a green boat. iii Find the name of sailors who have not reserved red boat iv Find the ids and names of sailors who have reserved two different boats on the same day. v Find the average age of sailors for each rating level that has at least two sailors.	10	L3	5
6.a.	Consider the relation scheme R = {E, F, G, H, I, J, K, L, M, N} and the set of functional dependencies {{E, F} -> {G}, {F} -> {I, J}, {E, H} -> {K, L}, K -> {M}, L -> {N} on R. What is the key for R? EFH.	4	L2	3
b..	Explain with an example Aggregate functions ,Grouping and Having clause in SQL	6	L2	1

Course Outcomes:

CO1: Understand and explore the needs and concepts of relational, NoSQL database and Distributed Architecture
CO2: Apply the knowledge of logical database design principles to real time issues.
CO3: Analyze and design data base systems using relational, NoSQL and Big Data concepts
CO4: Develop applications using relational and NoSQL database
CO5: Demonstrate database applications using various technologies.

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5
Total Marks	16	22	22	-	-	-	14	08	18	10	-